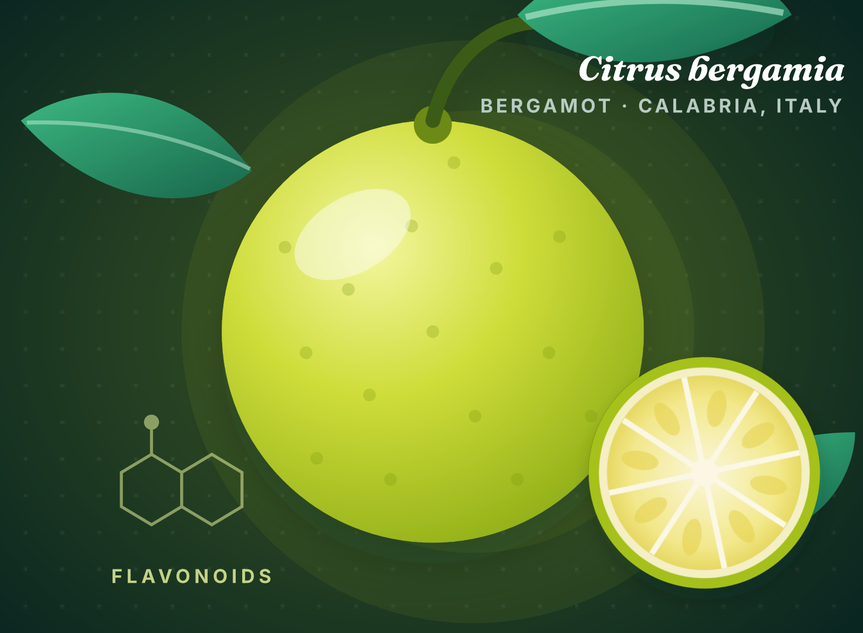
PHARMACOGENOMICS · CLINICAL PHARMACY · NUTRACEUTICALS

Locked *Out*

Genetics, statins & *Citrus bergamot* in familial hypercholesterolemia

Noor Al Ahmad • Clinical Pharmacy • Review & proposed study



THE CHOLESTEROL-LOWERING TOOLKIT

Statins
HMG-CoA reductase inhibitors
30–50%+
LDL-C LOWERING

Ezetimibe
NPC1L1 inhibitor · gut absorption
+≈20%
ADDED

Bempedoic acid
ATP-citrate lyase inhibitor
+15–25%
ADDED

PCSK9 antibodies
evolocumab · alirocumab
+50–60%
ADDED

Inclisiran
siRNA silencing hepatic PCSK9
+≈50%
ADDED

Evinacumab
ANGPTL3 antibody · receptor-free route
≈50%
IN HOMOZYGOUS FH

★ INVESTIGATIONAL ADJUNCT

Citrus bergamot
Bergamot polyphenol fraction · oral supplement
−40 to −55
MG/DL LDL-C · LOW-CERTAINTY RCTS

Approximate LDL-C reduction. Statins alone; others as add-on in pivotal trials (Cuchel 2023; ESC/EAS 2025). Bergamot: pooled small RCTs in dyslipidaemia, not FH-specific (Sadeghi-Dehsahraei 2022; Chambari 2026).

01 Spotlight: *Citrus bergamia*

A Mediterranean citrus whose polyphenols may act on the same pathway as statins.

MEET THE FRUIT

Bitter, green, and full of flavonoids

Calabria, Italy
Grown mainly on the Ionian coast

Rich in flavanones
Bergamot polyphenol fraction (BPF)

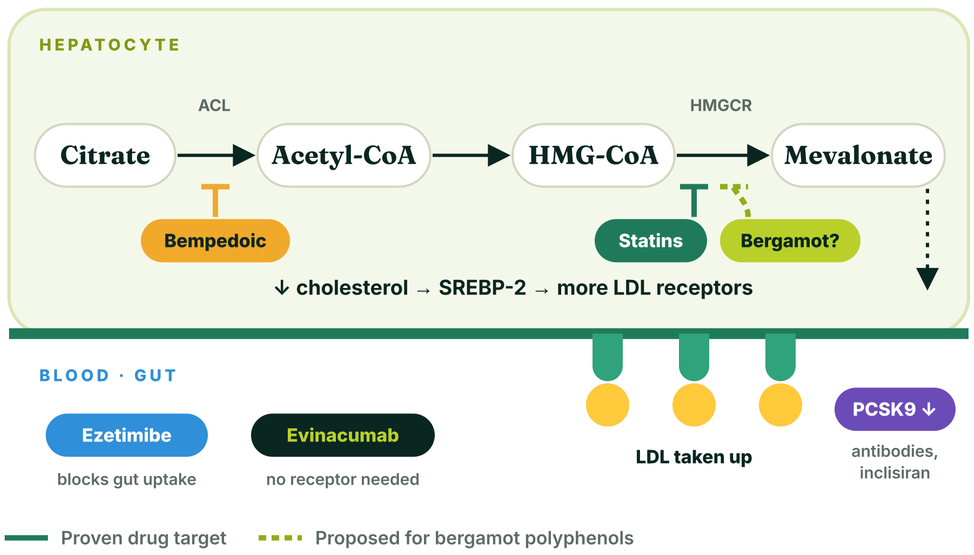
Taken as capsules
Standardised extract in trials

Naringin · Neohesperidin · Neoeriocitrin · Brutieridin* · Melitidin*

*HMG-flavanones with a statin-like side chain; proposed HMG-CoA reductase inhibitors.

WHERE EACH THERAPY ACTS

One pathway, many locks



WHAT THE TRIALS SAY

Promising numbers, weak evidence

−55 mg/dL — **LDL-C, 14 trials**
Meta-analysis, Phytother Res 2022

−40 mg/dL — **LDL-C, 9 RCTs · 615 adults**
Meta-analysis, Food Sci Nutr 2026

#1 — **of 10 nutraceuticals**
for LDL-C in a network meta-analysis of 131 trials

GRADE certainty: low to very low
VERY LOW · LOW · MODERATE · HIGH
6 of 9 trials poor quality · none in FH · **an adjunct, not a statin replacement**

02 Why one dose does not fit all

The FH gene sets how far LDL-C can fall; drug genes set how safely.

I The Gene
Half the gates are missing



LDLR	85–90%
APOB	5–10%
PCSK9	1–3%
LDLRAP1	<1%

No FH 2 working alleles		Normal LDL-C
Heterozygous 1 faulty allele		>190 MG/DL
Homozygous 2 faulty alleles		>400 MG/DL

Gates = working LDL receptors (schematic). ≈3× cardiovascular risk in carriers. Hu 2020; Cuchel 2023.

II The Drug
Same dose, different patients

Block HMG-CoA reductase · **Sense** SREBP-2 on · **Build** more receptors · **Clear** LDL-C ↓ 30–50%

SLCO1B1 C.521T>C · SIMVASTATIN MYOPATHY

4.5× odds per C allele

16.9× odds in CC vs TT

SLCO1B1 · *ABCG2* · *CYP2C9* CPIC 2022 genes

CPIC: switch statin or keep simvastatin <20 mg/day. Never a reason to stop statins.

SEARCH Collaborative Group, NEJM 2008; Cooper-DeHoff, Clin Pharmacol Ther 2022.

III The Patient
Where bergamot would fit



Four clues → one prescription: drug, dose, add-on, monitoring

PROPOSED RESEARCH QUESTION

Could genetically guided therapy plus diet and *Citrus bergamot* improve LDL-C control in young patients with FH?

- **Genotype** predicts how well statins work
- **Drug genes** predict how safely
- **Bergamot**: test it as an add-on, in FH

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Noor Al Ahmad
Pharmacogenomics · Clinical Pharmacy